Digital Interventions for Depression and Anxiety: A Systematic Synthesis of Chatbots, App-Based CBT, and Immersive VR/XR

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Abstract: This review synthesizes recent evidence on digital interventions for depression and anxiety, spanning chatbots, app-based cognitive behavioral therapy (CBT), and immersive VR/XR. Following PRISMA 2020, searches of PubMed and Web of Science (2021–2025) identified primary studies evaluating symptom change, feasibility, or engagement. Across modalities, randomized and quasi-experimental studies generally reported short-term reductions in depressive and anxiety symptoms relative to comparison conditions, with consistent signals for CBT-oriented chatbots and apps and additional improvements reported in VR/XR studies. Engagement varied widely and was supported by empathic tone, brief structured activities, light gamification, and contextual tailoring; repetitive or generic responses reduced adherence. Common limitations included modest samples, brief follow-up, reliance on self-report, variable control conditions, and inconsistently described safety and privacy practices. Overall, digital tools appear feasible, acceptable, and potentially effective short-term supports; priorities include larger and longer trials with diverse populations, clinician-verified outcomes, robust crisis-escalation and data-governance procedures, and evaluation of blended human–AI models.

1 INTRODUCTION

Mental health disorders, particularly depression and anxiety, affect over 580 million people worldwide and are a leading cause of disability (World Health Organization, 2022). The COVID-19 pandemic has made this crisis worse, with global prevalence increasing by about 25% (World Health Organization, 2023). Despite the availability of effective treatments such as cognitive behavioral therapy (CBT) and medication, nearly three-quarters of people with these conditions do not receive care. Barriers such as the shortage of trained professionals, high costs, limited access in rural areas, and persistent stigma continue to widen this treatment gap (Patel et al., 2018).

In recent years, technology has begun to play a more prominent role in mental health care. Digital tools including smartphone apps, teletherapy, and chatbots are increasingly being used to make psychological support more accessible. Among these, chatbots, which simulate conversation through text or voice, have drawn particular interest because they can deliver mental health support anytime and anywhere. These systems often use techniques from CBT, mindfulness, or motivational interviewing to help users cope with symptoms. Well-known examples such as Woebot, Wysa, and Tess have shown encouraging results in helping users manage depression and anxiety (Fitzpatrick et al., 2017).

Several randomized controlled trials (RCTs) have evaluated chatbot-based interventions and found reductions in depression and anxiety symptoms compared to control groups (Kahlon et al., 2021)(Fitzpatrick et al., 2017). Recent meta-analyses also suggest that AI-driven chatbots can produce small-to-moderate improvements in mood and well-being (Vaidyam et al., 2024). This indicates that chatbot interventions may be a practical, low-cost way to extend mental health support, especially for those unable or unwilling to access traditional therapy.

However, challenges remain. Some users find chatbot interactions too repetitive or impersonal, leading to reduced engagement over time. There are also ongoing concerns about privacy, data security, and the absence of human empathy. As conversational AI systems become more sophisticated, it is essential to continue evaluating their effectiveness and safety through rigorous research.

This review aims to synthesize randomized, quasi-experimental, and feasibility studies of

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digital interventions for depression and anxiety—including chatbots, app-based CBT, and immersive VR/XR—focusing on effectiveness, engagement, and key gaps for future research.

2 BACKGROUND

2.1 The Global Mental Health Crisis

Depression and anxiety affect more than 580 million people worldwide and remain leading causes of disability (World Health Organization, 2022). The COVID-19 pandemic further worsened this burden, raising global prevalence by roughly 25% (World Health Organization, 2023). Although evidence-based treatments such as cognitive behavioral therapy (CBT) and medication are available, nearly three-quarters of affected individuals receive no care. Contributing factors include the shortage of trained professionals, high treatment costs, limited rural access, and persistent stigma (Patel et al., 2018). Certain groups face especially high barriers, including university students balancing academic pressures (Liu et al., 2022; He et al., 2022), postpartum women (Danieli et al., 2022), and people with chronic illnesses who often experience comorbid mental-health symptoms (Karkosz et al., 2024).

2.2 Digital Mental Health Interventions and Chatbots

Digital technologies have expanded access to psychological support outside traditional therapy settings. Mobile apps, teletherapy, and AI chatbots allow users to receive guidance and coping tools at their own pace. Chatbots, in particular, simulate conversation through text or voice and often apply CBT, mindfulness, or psychoeducation principles to support users (Fitzpatrick et al., 2017). Well-known examples such as Woebot, Wysa, and Tess have shown promising results in reducing symptoms of depression and anxiety (Fitzpatrick et al., 2017; Liu et al., 2022).

2.3 Chatbot Design and Function

The design of mental-health chatbots varies based on purpose and complexity. Some rely on rule-based systems with preset decision paths (Tong et al., 2025), while others use large-language models to enable more natural, flexible dialogue (Heinz et al., 2025; Sabour et al., 2023). Many are fully self-guided (Karkosz et al., 2024; Liu et al., 2022), whereas others incorporate gamified elements to boost engagement (Bress et al., 2024; Kim et al., 2025) or act as adjuncts to professional therapy (MacNeill et al., 2024; Suharwardy et al., 2023). Newer approaches integrate immersive technologies such as virtual and extended reality to create more interactive, responsive mental-health experiences (Lee et al., 2024; Paul et al., 2024).

2.4 Scope and Rationale for this Review

Given the rapidly evolving landscape of digital mental health, this review intentionally broadens scope beyond chatbots alone to include app-based CBT programs and immersive technologies (VR/XR) that target depression and anxiety. We also consider studies focused on engagement, usability, and real-world usage (e.g., log analyses), and we do not restrict to adults when adolescent findings inform feasibility and design. This broader view allows us to synthesize effectiveness signals, characterize engagement patterns, and surface shared limitations and ethical considerations across modalities. The next section details our search, selection, and extraction procedures.

3 METHODS

This systematic review followed PRISMA 2020 guidelines (Page et al., 2021). The review question and eligibility criteria were defined a priori using a PICOS framework: Population (human participants with depressive and/or anxiety symptoms or populations where these outcomes were assessed), Intervention (patient-facing digital mental health interventions, including chatbots/conversational agents, app-based CBT, and immersive VR/XR tools), Comparator (when applicable: usual care, waitlist, active control, or alternative digital intervention), Outcomes (validated depression/anxiety symptom measures and/or feasibility, engagement, usability outcomes), and Study design (primary empirical studies). The protocol was not registered (e.g., PROSPERO/OSF); this is noted as a limitation.

3.1 Search Strategy

We searched PubMed and Web of Science for records published between January 1, 2021 and October 15, 2025. The final search was executed on October 15, 2025. Search strings combined terms for depression/anxiety and chatbots/conversational agents. No language restrictions were applied at the search stage.

The complete database queries are provided below:

PubMed:

```
((depress*[Title/Abstract] OR
anxi*[Title/Abstract] OR "generalized
anxiety"[Title/Abstract] OR "mood
disorder*"[Title/Abstract] OR
"mental health"[Title/Abstract]
OR PHQ-9[Title/Abstract]
OR GAD-7[Title/Abstract]
OR PHQ[Title/Abstract] OR
GAD[Title/Abstract]) OR
"Mental Disorders"[MeSH]) AND
(chatbot*[Title/Abstract] OR
"chat bot"[Title/Abstract] OR
"chat-bot"[Title/Abstract] OR
"conversational agent*"[Title/Abstract]
OR "conversational
AI"[Title/Abstract] OR "virtual
agent*"[Title/Abstract] OR "artificial
intelligence"[Title/Abstract] OR
AI[Title/Abstract])
```

Web of Science:

```
TS=((depress* OR anxi* OR "generalized
anxiety" OR "mood disorder*" OR
"mental health" OR PHQ-9 OR GAD-7 OR
PHQ OR GAD) AND (chatbot* OR "chat
bot" OR "conversational agent*"
OR "conversational AI" OR "virtual
agent*"))
```

3.2 Screening and Selection Process

Search results were exported and de-duplicated using automated and manual checks. Screening was conducted in two stages: (1) title/abstract screening and (2) full-text review. Titles/abstracts and full texts were screened against the eligibility and exclusion criteria described below. Three reviewers independently screened records at each stage. Disagreements were resolved through discussion and consensus; when needed, a fourth reviewer adjudicated.

The overall study selection process is summarized in the PRISMA flow diagram (Figure 1).

3.3 Eligibility and Exclusion Criteria

Studies were eligible if they reported primary empirical data on a patient-facing digital mental health intervention relevant to depression and/or anxiety. This encompassed chatbots (rule-based or large-language-model based), app-based CBT programs, and immersive VR/XR interventions. Eligible studies either enrolled participants with depressive and/or anxiety symptoms (clinical or subclinical) or evaluated feasibility, engagement, or usability outcomes in relevant populations.

For efficacy-focused studies, validated symptom measures were required (e.g., PHQ-9, BDI-II, GAD-7, HADS-A/D). For feasibility/usability-focused studies, studies were required to report engagement (e.g., adherence/usage) and/or usability/acceptability outcomes.

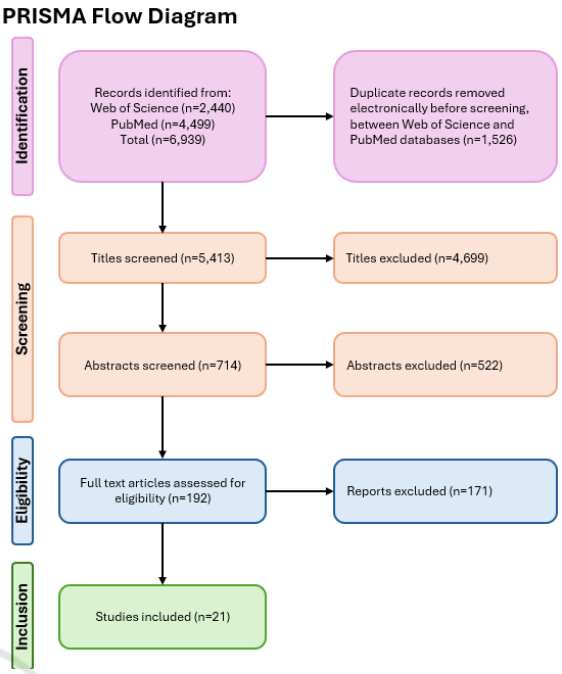


Figure 1: PRISMA flow diagram of study selection.

We excluded secondary research (reviews, meta-analyses, protocols, editorials), clinician-facing tools without a patient-facing component, and studies not related to depression or anxiety outcomes.

3.4 Data Extraction

Data were extracted using a standardized form. For each included study, we recorded bibliographic details (title, year), study design, population/sample characteristics, intervention modality and key features, comparator (if applicable), outcome measures, direction of findings (improvement/no change/worsening), engagement/adherence metrics, usability/acceptability outcomes, and any reported safety, privacy, or data governance features. Extracted fields also included author-stated limitations.

3.5 Quality Appraisal/ Risk of Bias

Study quality was assessed using the Mixed Methods Appraisal Tool (MMAT) (Hong et al., 2018). Two reviewers independently applied design-specific MMAT criteria, resolving disagreements by consensus (or a third reviewer). Appraisal informed interpretation of findings but did not determine inclusion.

4 RESULTS

4.1 Overview of Included Studies

This review examined 21 recent studies from 2021 to 2025 that explored different digital approaches to treating anxiety and depression. The studies covered a wide range of tools: some looked at fully automated chatbots like Woebot(Fitzpatrick et al., 2017), Fido(Karkosz et al., 2024), XiaoE(He et al., 2022), and Emohaa(Sabour et al., 2023) that guide users through cognitive behavioral therapy; others tested hybrid apps and mobile programs such as Maya(Bress et al., 2024) and Mind Booster Green(Kim et al., 2025). A few studies explored newer AI-powered conversational systems like Therabot(Heinz et al., 2025), while others investigated immersive technologies including virtual reality biofeedback(Cho et al., 2024), VR-based CBT, and extended reality behavioral activation. Some research focused specifically on whether these tools are engaging and easy to use studies like ChatPal(Booth et al., 2023), TEO(Danieli et al., 2022), and work on chatbots for teenagers fell into this category. Additionally, some research focused on specific populations that could benefit, such new moms or those with chronic illnesses(Suharwardy et al., 2023). All things considered, these research demonstrate how rapidly AI-based mental health care is developing and taking new turns.

4.2 Effectiveness of Interventions

Statistically substantial symptom reduction was found in the majority of randomized and quasi-experimental investigations. The CBT-based chatbots *Woebot*(Fitzpatrick et al., 2017) , *Fido* (Karkosz et al., 2024), and *XiaoE* (He et al., 2022) consistently decreased PHQ-9 and GAD-7 scores over two to six weeks. The *Emohaa* RCT (Sabour et al., 2023) achieved strong short-term improvements in mood and stress. App-based interventions such as *Maya* (Bress et al., 2024) and *Mind Booster Green* (Kim et al., 2025) produced clinically meaningful anxiety reductions among university students, while *Therabot* (Heinz et al., 2025), the only large-language-model chatbot, reported up to 51% reduction in depression and 31% in anxiety, paralleling therapist-led CBT outcomes. Immersive studies using VR-biofeedback (Cho et al., 2024), VR-CBT(Cho et al., 2024) , XR-BA , and hybrid VR therapy prototypes also demonstrated enhanced relaxation, emotion regulation, and motivation. Overall, effectiveness was strongest in structured CBT or mindfulness frameworks, though the duration of effect remains uncertain due to limited follow-up.

4.3 User Engagement and Acceptability

Engagement looked very different from one study to another. In trials such as ChatPal(Booth et al., 2023) and the adolescent chatbot project, people said they liked having a space where they could talk freely and whenever they needed to, without feeling judged. The Emohaa (Sabour et al., 2023) and Therabot(Heinz et al., 2025) studies found that users stayed longer when the chatbot sounded friendly and responded in a way that felt more personal. The Maya app(Bress et al., 2024) also saw better participation when small rewards or light, game-style suggestions were included. On the other hand, users of Fido(Karkosz et al., 2024) and XiaoE(He et al., 2022) often lost interest when the replies seemed repetitive or too mechanical. Work with older adults in TEO(Danieli et al., 2022) and with new mothers in the postpartum chatbot study(Suharwardy et al., 2023) showed that people were more likely to keep using the system when messages reflected their own age, culture, or life situation. Overall, these findings suggest that how users connect emotionally with a chatbot matters more than the technology itself in keeping them engaged.

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5 DISCUSSION

5.1 Methodological and Technological Limitations

Several methodological weaknesses were consistent across studies. Sample sizes were typically modest (30–300 participants) (Liu et al., 2022)(Bress et al., 2024)(Cho et al., 2024); intervention durations were short (1–8 weeks)(Liu et al., 2022)(Bress et al., 2024); and many relied solely on self-reported measures. Only a few investigations incorporated long-term follow-ups or clinician verification. The absence of active control groups and high dropout rates (up to 45%) reduced internal validity. Another important issue is that many key interventions such as Woebot, Fido, and XiaoE were tested primarily with young adults, yet this age bias was rarely acknowledged in the discussions. Technological barriers, such as constrained conversational logic in *Woebot*(Fitzpatrick et al., 2017), network dependence in Fido(Karkosz et al., 2024), and equipment requirements for VR and XR systems(Lee et al., 2024) restricted scalability. In

addition, several studies Topic-based Chatbots; AI for College Anxiety(Kim et al., 2025) noted that inconsistent language understanding and lack of emotional nuance reduced perceived trustworthiness. Collectively, these factors point to a research field still in early experimental stages.

5.2 Ethical and Practical Considerations

Across most of the reviewed studies, ethical concerns appeared repeatedly. None of the trials included real-time crisis response or referral options for users in distress. Privacy and data handling issues were seldom addressed, especially in generative-AI programs like *Therabot*(Heinz et al., 2025) and cross-platform tools like *Maya*(Bress et al., 2024). Studies on postpartum care(Suharwardy et al., 2023) and chronic illness management(MacNeill et al., 2024) found that some chatbot responses didn't always fit the situation. In a few sensitive cases, the automated replies felt too general or even confusing when users phrased questions in different ways. Several authors also brought up bias in the underlying algorithms, uneven digital skills among participants, and how much these systems rely on commercial AI software. Over time, it seems important for projects like this to have clearer data policies, real ethical oversight, and some form of clinician involvement to keep things reliable and safe.

5.3 Future Directions

When reviewing the twenty-one studies, it's clear that a wide mix of digital and AI tools has been tested from CBT chatbots and conversational bots to a few virtual-reality programs. In most cases, these tools helped lower anxiety and depression levels to some extent, and many participants said they liked using them. That said, the studies also ran into a few repeating problems, such as unclear research setups, concerns about data privacy, and the difficulty of keeping people engaged over time. Going forward, researchers might try running larger, longer studies that bring in participants from more varied backgrounds. It would also be helpful to pay closer attention to how outcomes are measured and to explore better ways of blending human involvement with AI-based support.

6 CONCLUSION

This review synthesized evidence across chatbots, app-based CBT, and immersive VR/XR for depression and anxiety. Across modalities, studies generally show short-term symptom improvements and favorable user attitudes, with engagement boosted by empathic tone, brief structured activities, and personalization. However, modest samples, short follow-up, reliance on self-report, variable controls, and inconsistent safety/privacy practices limit confidence and generalizability. Priorities include larger, longer, and more diverse trials, clearer outcome reporting (including clinician-verified measures), and robust crisis-escalation and data-governance procedures. With these advances, digital tools can be more reliably integrated into stepped-care pathways as accessible adjuncts to traditional services.

REFERENCES

Booth, F., Potts, C., Bond, R., Mulvenna, M., Kostenius, C., Dhanapala, I., Vakaloudis, A., Cahill, B., Kuosmanen, L., and Ennis, E. (2023). A mental health and well-being chatbot: User event log analysis. *JMIR mHealth and uHealth*, 11:e43052.

Bress, J. N., Falk, A., Schier, M. M., et al. (2024). Efficacy of a mobile app-based intervention for young adults with anxiety disorders: A randomized clinical trial. *JAMA Network Open*, 7(8):e2428372.

Cho, Y., Kim, H., Seong, S., Park, K., Choi, J., Kim, M., Kim, D., and Jeon, H. J. (2024). Effect of virtual reality-based biofeedback for depressive and anxiety symptoms: Randomized controlled study. *Journal of Affective Disorders*, 361:392–398. Accessed October 20, 2025.

Danieli, M., Ciulli, T., Mousavi, S., Silvestri, G., Barbato, S., Di Natale, L., and Riccardi, G. (2022). Assessing the impact of conversational artificial intelligence in the treatment of stress and anxiety in aging adults: Randomized controlled trial. *JMIR Mental Health*, 9(9):e38067.

Fitzpatrick, K. K., Darcy, A., and Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (woebot): A randomized controlled trial. *JMIR Mental Health*, 4(2):e19.

He, Y., Yang, L., Zhu, X., Wu, B., Zhang, S., Qian, C., and Tian, T. (2022). Mental health chatbot for young adults with depressive symptoms during the covid-19 pandemic: Single-blind, three-arm randomized controlled trial. *Journal of Medical Internet Research*, 24(11):e40719.

Heinz, M. V., Mackin, D. M., Trudeau, B. M., Bhattacharya, S., Wang, Y., Banta, H. A., Jewett, A. D., Salzhauer, A. J., Griffin, T. Z., and Jacobson, N. C. (2025). Randomized trial of a generative ai chatbot for mental health treatment. *NEJM AI*, 2(4). ClinicalTrials.gov number: NCT06013137.

Hong, Q. N., Pluye, P., Fàbregues, S., et al. (2018). Mixed methods appraisal tool (mmat), version 2018. User guide.

Kahlon, S., Lindner, P., and Nordgreen, T. (2021). Chatbot-based cognitive behavioral therapy for depression, anxiety, and stress: Randomized controlled trial. *Frontiers in Digital Health*, 3:721398.

Karkosz, S., Szymański, R., Sanna, K., and Michałowski, J. (2024). Effectiveness of a web-based and mobile therapy chatbot on anxiety and depressive symptoms in subclinical young adults: Randomized controlled trial. *JMIR Formative Research*, 8:e47960. Accessed October 20, 2025.

Kim, M. H., Moon, Y. K., and Chung, K. M. (2025). An app-based cognitive behavioral therapy program tailored for college students: Randomized controlled trial. *JMIR mHealth and uHealth*, 13:e50006.

Lee, M., Jang, S., Shin, H. K., Choi, S.-W., Kim, H. T., Oh, J., Kwon, J. H., Choi, Y., Kang, S., Back, I.-S., et al. (2024). Virtual reality-based cognitive behavior therapy for major depressive disorder: an alternative to pharmacotherapy for reducing suicidality. *Yonsei Medical Journal*, 66(1):25.

Liu, H., Peng, H., Song, X., Xu, C., and Zhang, M. (2022). Using ai chatbots to provide self-help depression interventions for university students: A randomized trial of effectiveness. *Internet Interventions*, 27:100495. Accessed October 20, 2025.

MacNeill, A. L., Doucet, S., and Luke, A. (2024). Effectiveness of a mental health chatbot for people with chronic diseases: randomized controlled trial. *JMIR Formative Research*, 8:e50025.

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., Stewart, L. A., Thomas, J., Tricco, A. C., Welch, V. A., Whiting, P., and Moher, D. (2021). The prisma 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71.

Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., and Unützer, J. (2018). The lancet commission on global mental health and sustainable development. *The Lancet*, 392(10157):1553–1598.

Paul, M., Bullock, K., Bailenson, J., Burns, D., et al. (2024). Examining the efficacy of extended reality–enhanced behavioral activation for adults with major depressive disorder: Randomized controlled trial. *JMIR Mental Health*, 11(1):e52326.

Sabour, S., Zhang, W., Xiao, X., Zhang, Y., Zheng, Y., Wen, J., Zhao, J., and Huang, M. (2023). A chatbot for mental health support: Exploring the impact of emohaa on reducing mental distress in china. *Frontiers in Digital Health*, 5:1133987. Accessed October 20, 2025.

Suharwardy, S., Ramachandran, M., Leonard, S. A., Gunaseelan, A., Lyell, D. J., Darcy, A., Robinson, A., and Judy, A. (2023). Feasibility and impact of a mental health chatbot on postpartum mental health: A randomized controlled trial. *AJOG Global Reports*, 3(3):100165. Accessed October 20, 2025.

Tong, A. C., Wong, K. T., Chung, W. W., and Mak, W. W. (2025). Effectiveness of topic-based chatbots on mental health self-care and mental well-being: Randomized controlled trial. *Journal of medical Internet research*, 27:e70436.

Vaidyam, A. N., Halamka, J., and Torous, J. (2024). Conversational agents in mental health: Systematic review and meta-analysis. *npj Digital Medicine*, 7(1):64.

World Health Organization (2022). Covid-19 pandemic triggers 25% increase in prevalence of anxiety and depression worldwide. Available from: https://www.who.int/news/item/02-03-2022-covid-19-pandemic-triggers-25-increase-in-prevalence-of-anxiety-and-depression-worldwide. Accessed: 15 October 2025.

World Health Organization (2023). Depression and other common mental disorders: Global health estimates.

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